

Pratyush Agrawal

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Education

Manipal Institute of Technology,
B.Tech in Information Technology – Karnataka, India

2024 – 2028
CGPA: 7.86 / 10

Coursework

Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Discrete Mathematics, Computer Organization

Projects

KV Cache Quantization | *Python, PyTorch, Hugging Face Transformers, LLM Optimization* [GitHub](#) | [Blog](#)

- Reduced GPT-2 KV cache footprint by **68.5%** with only a **0.71% increase in perplexity** using a KIVI-inspired INT8 quantization pipeline.
- Architected a custom inference engine in PyTorch, bypassing `model.generate()` to implement manual cache management and token-by-token generation for granular control over tensor states.
- Engineered flexible quantization schemes supporting symmetric/asymmetric modes across whole-tensor, per-channel, and per-token granularities to evaluate reconstruction error on real GPT-2 tensors.
- Conducted comprehensive benchmarking of FP32, FP16, and INT8 formats, analyzing the trade-offs between inference latency, memory consumption, and teacher-forced perplexity.

DataTalk AI | *React.js, Node.js, Express.js, Azure OpenAI, Azure SQL Database*

[GitHub](#) | [Demo](#)

- Built a multilingual (English/Hindi/Bengali) natural-language BI dashboard letting retailers upload raw CSV sales data and ask plain-language questions, translating them into validated T-SQL executed against Azure SQL Database.
- Designed a dual-engine AI architecture routing SQL generation and insight generation to GPT-4o and fast intent extraction to o4-mini, balancing analytical depth against response latency.
- Implemented SQL execution guardrails – SELECT-only enforcement, blocked DDL/DML keywords, and parameterized queries – to safely run AI-generated queries against live retail data.
- Extended the same NL-to-SQL pipeline into quick-report presets and dedicated profit/ledger views (top products, regional revenue, monthly trends).

Achievements

Microsoft AI Unlocked Hackathon

2025

- Ranked in the top 50 out of 300+ teams nationwide with DataTalk AI

Skills

Languages: Python, Java, JavaScript, SQL

Libraries & Frameworks: Django, React.js, Node.js, Express.js, PyTorch

Tools & Platforms: Git, GitHub, Azure, PostgreSQL, Render

Extra Curricular

IEEE Student Branch, Manipal Institute of Technology

2024 – Present

- Managing Committee Member and WebDev Domain POC for TechWeek-IEEE SBM's flagship technical fest
- Co-organized DevRush hackathon with 18 participating teams